Algebra II with Trigonometry Chapter 6.1

Study Guide (Gemini)

Saved tutoring response with MathJax rendering. Original PDF file: 4506904_alg-2trig-h-chp-6-1-note-docx.pdf

Here is a comprehensive study guide and tutoring response based on the provided unit circle worksheet.

1. Short Summary

This document is a guided worksheet for an Algebra 2/Trigonometry Honors course introducing the concept of the Unit Circle. It covers the equation of the unit circle, how to locate "terminal points" determined by an arc length t, and how to calculate "reference numbers" to figure out the coordinates of points in different quadrants. It transitions from using algebraic proofs (verifying points via the circle's equation) to understanding the geometric symmetry of the circle.

2. Core Definitions, Theorems, and Formulas

- **The Unit Circle:** A circle with a radius of 1 centered at the origin $(0, 0)$ in the xy-plane.
- **Equation of the Unit Circle:** $x^2 + y^2 = 1$
- **Circumference Formula:** $C = 2\pi$
- **Terminal Point $P(x, y)$:** The specific (x, y) coordinate on the unit circle you arrive at after traveling a distance t along the edge of the circle, starting from $(1, 0)$.
- **Reference Number ($\bar{t}$):** The shortest distance along the unit circle between the terminal point determined by t and the x-axis. It is always a positive number between 0 and $\frac{\pi}{2}$.
- **Coterminal Rule:** Adding or subtracting 2π (one full revolution) to a value t does not change the terminal point.

3. Worked Study Guide (Step-by-Step)

Since the document contains fill-in-the-blanks, here is a guide that resolves the missing information and explains the concepts step-by-step.

Step 1: Understanding the Unit Circle Basics (Page 1)

- The unit circle is the circle of **radius 1** centered at the **origin** $(0, 0)$.
- Starting at the point $(1, 0)$, moving in a **counterclockwise** direction means t is positive. Moving in a **clockwise** direction means t is negative.
- Because the radius $r = 1$, the circumference is $2\pi(1) = \mathbf{2}\pi$.
- Halfway around the circle is π. A quarter of the way around is $\frac{\pi}{\mathbf{2}}$.

Step 2: Proving a Point is on the Unit Circle (Example 1) To prove a point is on the unit circle, plug the x and y coordinates into $x^2 + y^2 = 1$.

- *Example 1a:* $(-\frac{5}{7})^2 + (-\frac{2\sqrt{6}}{7})^2 = \frac{25}{49} + \frac{4(6)}{49} = \frac{25}{49} + \frac{24}{49} = \frac{49}{49} = 1$. Because it equals 1, the point is on the circle.

Step 3: Finding Missing Coordinates (Examples 2 & 3) If you know one coordinate and the quadrant, you can find the other.

- *Method:* Plug the known coordinate into $x^2 + y^2 = 1$ and solve for the unknown.
- *Applying Signs:*
- Quadrant I: x is (+), y is (+)
- Quadrant II: x is (-), y is (+)
- Quadrant III: x is (-), y is (-)
- Quadrant IV: x is (+), y is (-)
- *Look at Ex 2a:* $P(-\frac{3}{5}, y)$ in Quadrant III.

$(-\frac{3}{5})^2 + y^2 = 1 \Rightarrow \frac{9}{25} + y^2 = 1 \Rightarrow y^2 = \frac{16}{25}$. Therefore, $y = \pm\frac{4}{5}$. Since it is in QIII, y must be negative: $y = -\frac{4}{5}$.

Step 4: Finding Reference Numbers (Page 2 & Exercise 5) The reference number $\bar{t}$ is the shortest distance to the x-axis. Here is how to fill in the blanks for the quadrants:

- **Quadrant I:** $\bar{t} = t$

- **Quadrant II:** $\bar{t} = \pi - t$
- **Quadrant III:** $\bar{t} = t - \pi$
- **Quadrant IV:** $\bar{t} = 2\pi - t$

(Note: If t is negative, or greater than 2π, first add/subtract 2π until your value is between 0 and 2π.)

- *Dealing with integers (Ex 5c: $t = 6$):* Remember $\pi \approx 3.14$, so $2\pi \approx 6.28$. The value $t = 6$ is in Quadrant IV (just short of 6.28). The shortest distance to the x-axis is $2\pi - 6$.

Step 5: Using Symmetry (Example 8) If you know the terminal point for t is $P(x, y) = (\frac{3}{5}, \frac{4}{5})$:

- $-t$ flips the point over the x-axis: $(x, -y) \to (\frac{3}{5}, -\frac{4}{5})$
- $\pi - t$ flips the point over the y-axis (to QII): $(-x, y) \to (-\frac{3}{5}, \frac{4}{5})$
- $2\pi + t$ is a full circle plus t, so you land exactly where you started: $(x, y) \to (\frac{3}{5}, \frac{4}{5})$

4. Practice Problems

These problems require only the formulas, definitions, and logic established in the provided document.

Problem 1: Show that the point $P(\frac{\sqrt{3}}{2}, -\frac{1}{2})$ is on the unit circle. **Problem 2:** The point P is on the unit circle. The x-coordinate of P is $\frac{5}{13}$, and P lies in Quadrant IV. Find the y-coordinate. **Problem 3:** Find the reference number $\bar{t}$ for $t = \frac{7\pi}{6}$. **Problem 4:** Find the terminal point $P(x, y)$ determined by $t = -\pi$. **Problem 5:** Suppose the terminal point determined by t is $P(-\frac{4}{5}, \frac{3}{5})$. What is the terminal point determined by $t + 2\pi$?

Answers to Practice Problems

1. Use the equation $x^2 + y^2 = 1$. $(\frac{\sqrt{3}}{2})^2 + (-\frac{1}{2})^2 = \frac{3}{4} + \frac{1}{4} = \frac{4}{4} = 1$. The point is on the circle. **2.** $(\frac{5}{13})^2 + y^2 = 1 \Rightarrow \frac{25}{169} + y^2 = \frac{169}{169} \Rightarrow y^2 = \frac{144}{169} \Rightarrow y = \pm\frac{12}{13}$. Because P is in Quadrant IV, the y-coordinate must be negative. **Answer:** $y = -\frac{12}{13}$. **3.** The value $\frac{7\pi}{6}$ is slightly larger than π ($\frac{6\pi}{6}$), which places it in Quadrant III.

Using the Quadrant III rule: $\bar{t} = t - \pi \Rightarrow \bar{t} = \frac{7\pi}{6} - \frac{6\pi}{6} = \frac{\pi}{6}$. **Answer:** $\bar{t} = \frac{\pi}{6}$. **4.** Starting at $(1, 0)$ and moving clockwise by π (halfway around the circle) lands on the negative x-axis. **Answer:** $(-1, 0)$. **5.** Adding 2π represents traveling one full circumference around the unit circle, meaning you end up exactly at the same point you started. **Answer:** $(-\frac{4}{5}, \frac{3}{5})$.

5. Assumptions and Ambiguities Resolved

- **Fill-in-the-blanks:** The PDF is a student worksheet with many missing words (e.g., "centered at the _____"). I assumed standard trigonometric definitions to fill these in for the study guide (e.g., "origin", "counterclockwise").
- **Special Terminal Points (Page 3):** The document provides a blank 16-point unit circle (Example 4) but does not contain text explicitly calculating the $\frac{\pi}{6}$, $\frac{\pi}{4}$, and $\frac{\pi}{3}$ coordinate families within this specific document. I assumed the student is expected to memorize these or fill them out in class, so I focused the guide on the *algebraic and geometric methods* explicitly detailed in the text rather than providing a pre-filled 16-point circle diagram.
- **Non-Radian / Integer t values (Ex 5 & 7):** For instances like $t = 6$, $t = -7$, or $t = 2.5$, the document assumes the student knows to approximate $\pi \approx 3.14$ to locate the quadrant. I explicitly stated this assumption in Step 4 of the study guide to make it clear for the student.